ZIJIAN CAO / UNDERGRADUATE FINAL PROJECT

Comparing NACA aerofoils at low Reynolds numbers

Project report: April 2026 | Portfolio summary: October 2026

I compared NACA 2412 and NACA 4412 to examine lift, drag and pitching-moment trade-offs relevant to small fixed-wing UAV aerofoil selection.

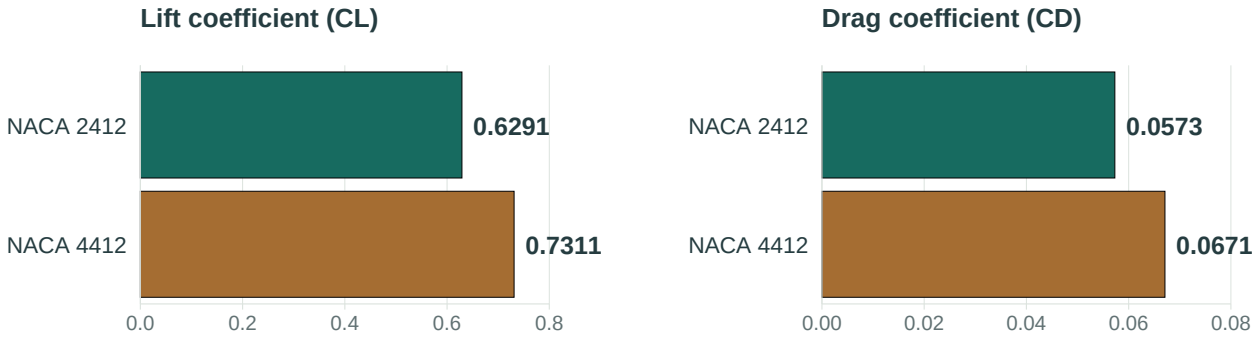
My work and methods

XFOIL comparison
Main numerical comparison at Reynolds numbers of 200,000 and 300,000; inspection of aerodynamic coefficients and their trends.

Complementary ANSYS Fluent cases
Two-dimensional cases at Reynolds numbers of 100,000, 200,000 and 300,000, at 0 and 8 degrees. SST k-omega, without a transition model.

Representative CFD comparison

Re = 200,000 | Angle of attack = 8 degrees



Source: Table 4.2, report p. 23. Separate coefficient scales; numerical CFD results.

Selected CFD entries replotted from Table 4.2, report p. 23. Both aerofoils are compared at the same Reynolds number and angle of attack. Lift and drag use separate coefficient scales.

Findings and engineering judgement

At this selected case, NACA 4412 has higher lift and drag coefficients than NACA 2412. This illustrates a design trade-off; it does not establish an overall optimum. My wider study compared aerodynamic coefficients, pitching-moment response and flow-field outputs, with attention to modelling limits.

Scope

This was a two-dimensional numerical study. It did not include wind-tunnel or flight testing, and the results do not establish full-aircraft performance. The Fluent cases provide a complementary numerical comparison, rather than experimental validation.

Source and preparation

Zijian Cao (April 2026), **Comparative Aerodynamic Study of Two NACA Aerofoil Profiles at Low Reynolds Numbers**. Methods: report pp. 11-18; selected coefficients: Table 4.2, p. 23; limitations and conclusions: pp. 30-38. This is a portfolio summary of the supplied dissertation.

OpenAI Codex assisted with drafting, formatting and replotting the stated table values on 25-27 October 2026. No additional simulation or experiment was performed for this summary.